1. What AI Is (and Isn't)

Course introduction, definitions, and the big picture

Week 1 · CE/ARCE 2XX AI Literacy for Civil and Architectural Engineering



COLLEGE OF ENGINEERING
Civil & Architectural Engineering & Mechanics

Agenda

Today's plan

1. Welcome: why this course, and who we are
2. Three AI applications in CE and ARCE — before any definition
3. Defining AI: why it is hard, and a definition that works
4. Talking about AI precisely (quiz: Yes / No / Kind of)
5. The big-picture map of AI
6. How this course works: format, grading, AI use
7. This week's lab day: your Python on-ramp

01

Welcome

Why an AI course for civil and architectural engineers?

Welcome

Why this course?

1 AI is already in your field

Drones find cracks, software forecasts building energy, contractors track sites with cameras, and chat assistants draft reports.

2 Employers expect AI literacy

Interns and new engineers are asked to use AI tools well — and to know when not to trust them.

3 Engineers stay responsible

A licensed engineer signs the drawings, not the software. You must be able to check what AI produces.

Welcome

Who I am

Dr. Liang Zhang

- Civil & Architectural Engineering & Mechanics (CAEM)
- Office: Civil Engineering 220F
- Email: liangzhang1@arizona.edu
- Office hours: see the syllabus

What I work on

- AI and data for buildings and infrastructure
- Turning sensor and simulation data into better design and operation decisions
- Teaching engineers to use AI with judgment

Welcome

Who are you, and how do you use AI?

4 min

Think · Pair · Share

- Answer the two poll questions on your phone or on paper
- Turn to a neighbor: share **one** time an AI tool helped you and **one** time it was wrong
- A few pairs share with the class

Poll: (1) How often do you use AI tools — never, monthly, weekly, daily? (2) For what — homework help, writing, coding, images, other?

02

What is AI?

Three examples before the definition

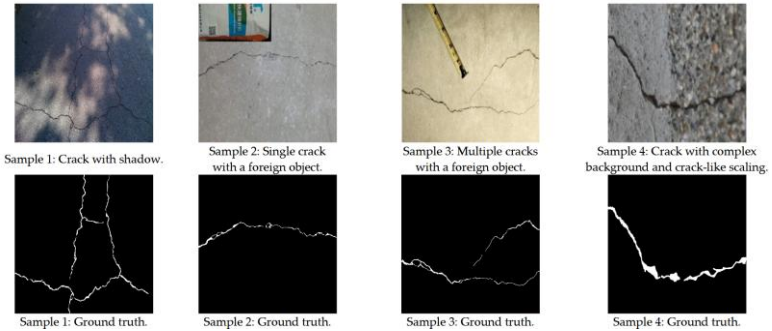
Three examples before the definition

- **Artificial intelligence**, or AI, means different things to different people, at different times, for different tasks
- Before we define it, we will look at three real applications in civil and architectural engineering
- Hold one question in mind: **what do these three have in common?**

Application 1

Image processing for inspection

- **Recognition:** AI finds cracks, corrosion, spalling, and thermal bridges in photos from drones or phones
- **Implications:** faster inspections, earlier warnings, safer inspectors (fewer people on ropes or lifts)
- **CE tie-in:** structural and infrastructure asset management
- **Same technology elsewhere:** face recognition, obstacle detection, counting wildlife

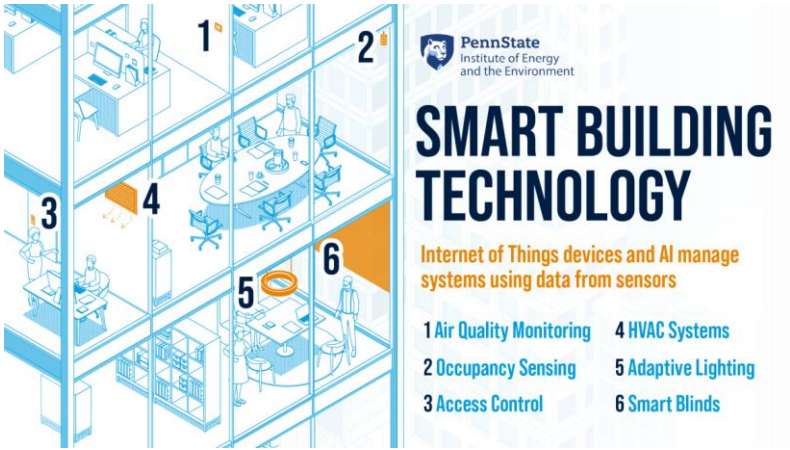


Crack photos and the cracks a model detected

Application 2

Smart buildings and energy management

- AI turns building operation data into decisions: **forecast** demand, **optimize** HVAC and lighting, **detect** faults early, **learn** occupancy patterns
- **Implications:** energy savings, comfort, decarbonization
- **ARCE tie-in:** a core architectural engineering area
- **Technique cluster:** prediction + optimization + decisions under uncertainty

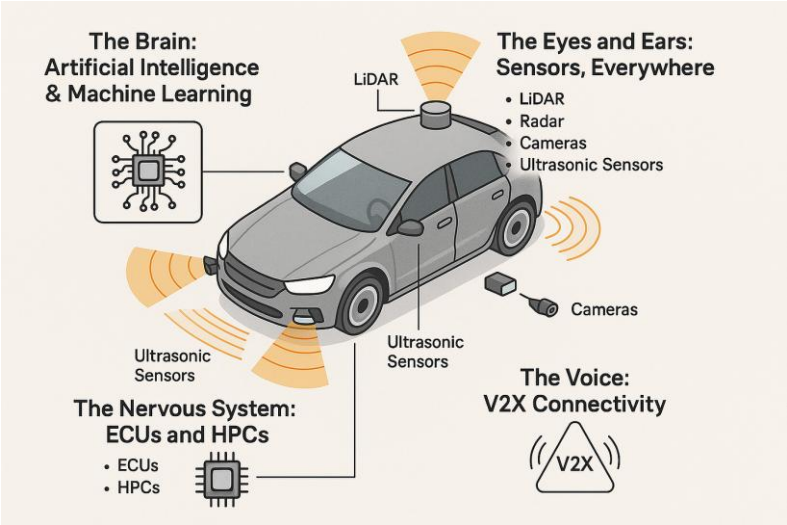


Smart building systems (Penn State IEE)

Application 3

Self-driving vehicles

- A combination of AI techniques: **search and planning**, **computer vision**, **control under uncertainty** — to cope with a complex, changing environment
- **Implications:** potential safety gains if reliability exceeds human drivers; more efficient logistics
- **CE tie-in:** transportation engineering — roads, signals, and traffic will change
- **Same technology:** delivery robots, drones, autonomous ships



Camera and lidar perception in an autonomous vehicle

What do the three have in common?

3 min

Think · Pair · Share

- With a neighbor, list what the three applications share
- Think about: **inputs** (what they sense), **outputs** (what they decide or do), and **how they got good**
- Be ready to share one word or phrase

Crack detection · Smart building control · Self-driving vehicle → ?

03

Defining AI

Why it is hard, and a definition that works

Defining AI

No single agreed definition

COINED "AI" IN 1955

John McCarthy

"The science and engineering of making intelligent machines." (2007)

TEXTBOOK

Russell & Norvig (1995)

An agent that perceives its environment and takes actions that maximize its chance of achieving its goals.

DICTIONARY

Oxford Languages

Computer systems able to perform tasks that normally require human intelligence.

POLICY

European Commission (2018)

Systems that show intelligent behavior by analyzing their environment and taking actions — with some autonomy — to achieve goals.

Defining AI

Four ways to think about AI

	Like a human	Rationally (ideally)
Thinking	**Thinking humanly** Cognitive modeling	**Thinking rationally** Laws of thought (logic)
Acting	**Acting humanly** The Turing Test (1950)	**Acting rationally** Rational agents — most engineering AI

A more useful definition: autonomy + adaptivity

Two properties of AI

Autonomy — the ability to perform tasks in complex environments without constant guidance by a user.

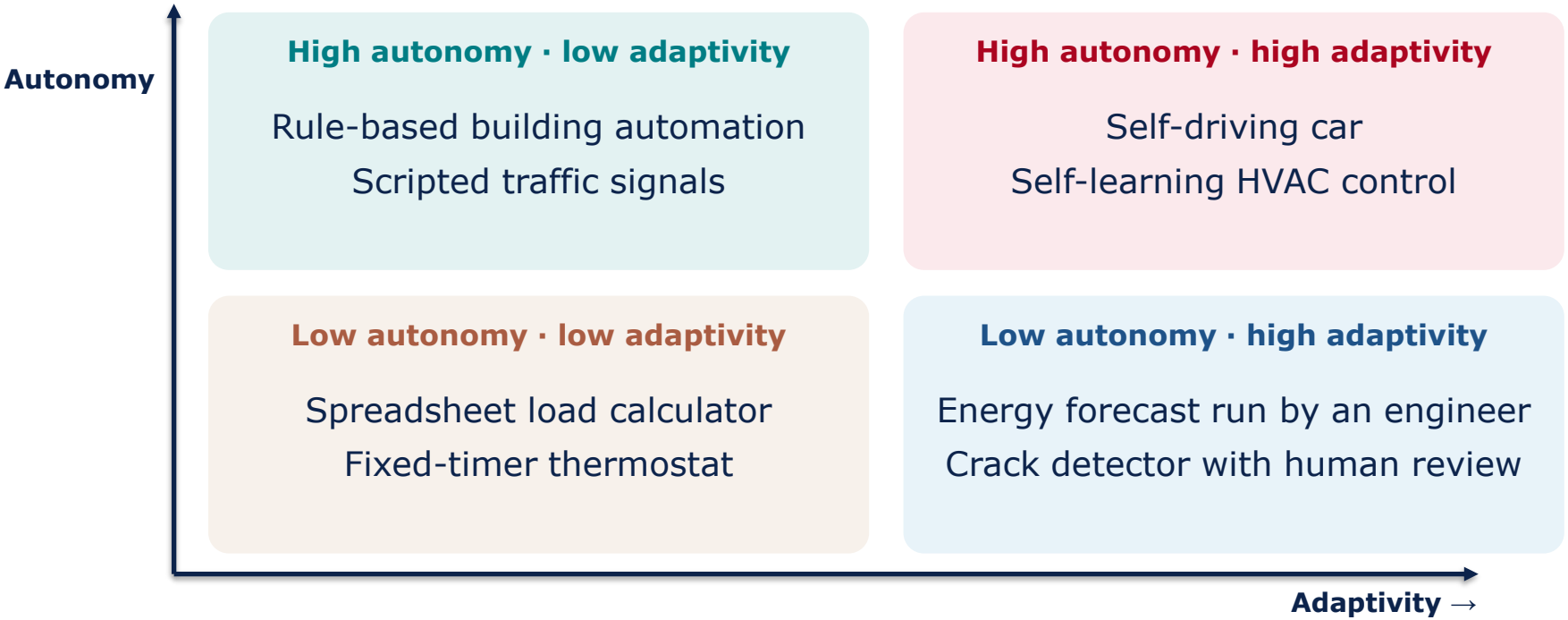
Adaptivity — the ability to improve performance by learning from experience.

Elements of AI, University of Helsinki & MinnaLearn

- Rather than one rigid definition: the more a system shows **both**, the more it counts as AI

Defining AI

The autonomy–adaptivity map



“A pinch of AI” — and how to talk about it

- AI is not a clean yes/no. Many methods have just “a pinch of AI,” like a pinch of salt — think in terms of **“AI-ness,”** not a hard line
- AI is a **discipline**, like mathematics or biology — not a countable object
- So don't say “an AI”; say **“an AI method”** or **“an AI system”** (you would never say “a biology”)
- Speaking about AI precisely is the start of **AI literacy** — a core goal of this course

Quiz

Yes, No, or Kind of? Is it AI?

#	System	Your answer
1	A spreadsheet that sums member weights to get total structural load	
2	Predicting a building's future energy use by fitting a curve to past consumption data	
3	A navigation app that finds the fastest route across a city's road network	
4	A building system that learns occupants' comfort preferences from past thermostat adjustments	
5	A cloud database that stores years of sensor data and streams it to many dashboards	
6	Adjusting brightness and contrast of a drone photo with predefined filters	
7	A deep-learning model that automatically detects and highlights cracks in every inspection photo	

Quiz

Answers — and the reasoning that matters

#	Answer	Why
1	No	Fixed formulas; no learning, no autonomy
2	Kind of	Learns from data (curve fitting = simple machine learning), but no autonomy
3	Kind of	Classic AI search and planning; acts on your behalf, but does not learn
4	Yes	Learns from experience and adjusts the system on its own
5	No	Important infrastructure for AI, but it only stores and moves data
6	No	Predefined filters; nothing is learned
7	Yes	Learned from labeled images; works without a person drawing each crack

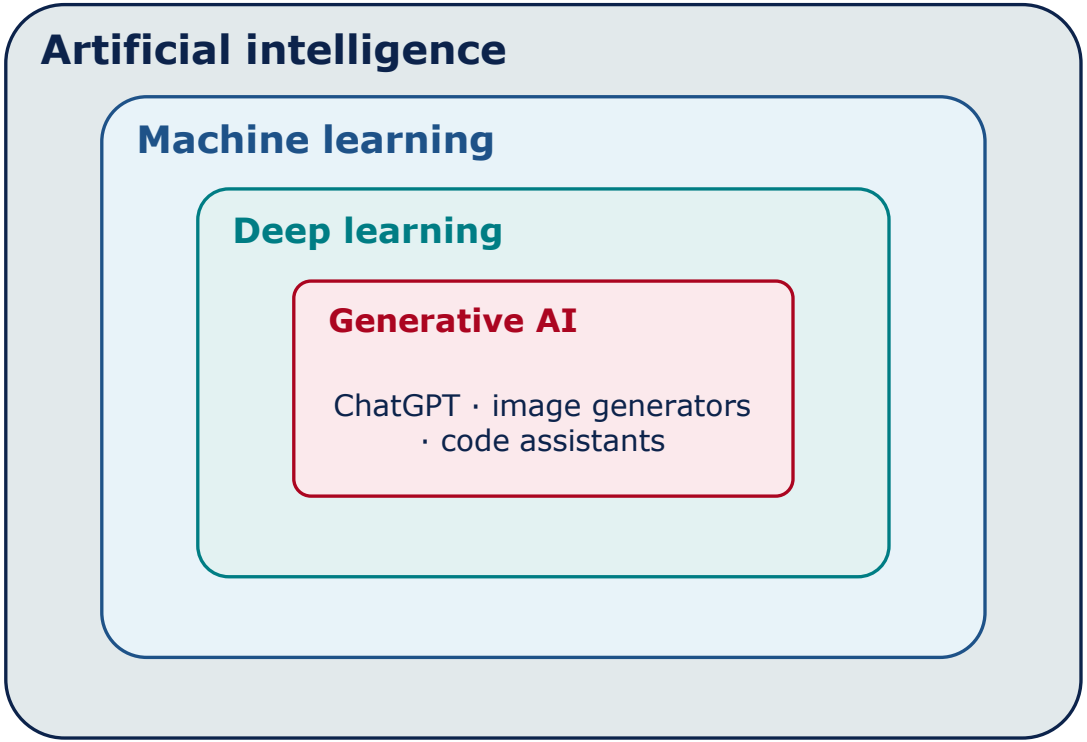
04

The big picture

A map of AI we will use all semester

The big picture

The nested map: AI ⊃ ML ⊃ DL ⊃ GenAI



AI beyond machine learning:

Search & planning
route finding, scheduling

Logic & rules
expert systems, code checking

Optimization
design optimization, dispatch

Robotics & control
drones, site robots

The big picture

A very short history of AI



- Turing's test (1950)
- "AI" coined in the 1955 Dartmouth proposal; workshop in 1956

- Logic, search, expert systems
- Two "AI winters" of lost funding

- Machine learning grows
- Deep Blue beats Kasparov (1997)

- Deep learning: images (2012), Go (2016)
- Transformer (2017)

- ChatGPT (2022) brings GenAI to everyone
- AI agents and AI policy

Myths

What AI is not

Myth	Reality
• AI understands the world the way people do	• It finds patterns in data; it can be fluent without understanding
• AI is objective because it is math	• It inherits the gaps and biases of its data and its designers
• If it sounds confident, it is right	• Generative AI can be confidently wrong (“hallucination”)
• AI will replace engineers	• It changes engineering tasks; engineers stay responsible for safety

vs

AI literacy = know how it works, use it well, verify it, and use it responsibly

05

This course

What we will cover, and how it works

This course

The course map

WEEKS 1–2

Foundations

- What AI is
- Machine learning, deep learning, generative AI
- Python and data

WEEKS 3–6

How machines learn

- Linear regression, cost, gradient descent
- Neural networks
- CNNs and sequence models

WEEKS 7–10

Generative AI and agents

- How LLMs work
- Prompting and retrieval (RAG)
- AI agents

WEEKS 11–15

AI in practice

- Applications across CE and ARCE
- Choosing and verifying AI tools
- Responsible AI, policy, ethics

This course

What this course is — and is not

This course IS

- AI literacy for engineers: concepts, hands-on use, judgment
- Hands-on Python in Google Colab — no installation, no prior coding
- Real CE and ARCE data: concrete, buildings, bridges, traffic
- Verification, responsible use, ethics, and policy

This course is NOT

- A computer-science course in AI theory or heavy math
- A software-engineering or app-building course
- A tour of every AI product on the market
- A course where AI does the work for you

This course

A typical week

Lecture day

- Concepts with CE/ARCE examples
- Short quiz on last week
- In-class activities

Lab day

- Guided Colab notebook
- Bring a laptop
- Exit ticket at the end

Homework

- About every two weeks
- Individual work
- Due Sunday 11:59 pm

Project

- Teams of 3–4 from Week 10
- Present in finals week

This course

How you are graded

Component	Weight	Notes
Computer labs	25%	Completion + exit ticket; lowest lab dropped
Homework (5)	20%	Individual; lowest homework dropped
Midterm exam	20%	Week 7 lab day; closed book; formula sheet provided
Final project	25%	Team application of AI to a CE/ARCE problem + presentation
Participation & quizzes	10%	Short concept quizzes and activities; lowest two dropped

This course

Using AI in this course: three labels

Every assignment shows its label. When in doubt, ask.

AI-Free

Do not use AI to produce what you submit. Example: quizzes, exam, exit tickets, some homework parts.

AI-Assisted

AI help is allowed (explaining, debugging, brainstorming). Add an AI-use acknowledgment. You own every line.

AI-Required

Using AI is part of the task — e.g., testing an LLM's answers. Document your prompts and outputs.

This course

Where everything lives

1 GitHub (course home)

Slides, lab notebooks, homework, project guide, datasets. Link on Brightspace and on the last slide.

2 Google Colab

Run every lab in your browser. Free; needs a Google account. Open notebooks from the GitHub links.

3 Brightspace (D2L)

Submit work, see grades and announcements, take quizzes.

Lab 01

Lab day: Python on-ramp

LAB

01

Bring a charged laptop

Open the notebook from the course GitHub page

- Open Google Colab and run your first notebook
- Python basics: variables, lists, loops, functions — just enough for this course
- **Activity 1:** run a pretrained vision model on concrete photos — can it tell cracked from intact?
- **Activity 2:** run a small language model from Python — and catch it making a mistake
- Exit ticket: three short questions

Wrap-up

Key takeaways

1. AI is already used across the life cycle of infrastructure and buildings
2. There is no single definition; use **autonomy + adaptivity** as a lens, and think in degrees
3. Say “an AI method,” and be specific about what it learned from and what it does
4. AI ⊃ machine learning ⊃ deep learning ⊃ generative AI — plus search, logic, optimization, robotics
5. AI literacy = understand it, use it, **verify it**, and use it responsibly

Learn more

Readings and resources

- **Elements of AI (free online course)** — Chapter 1 — read before next week
- **Course GitHub: github.com/TensorBuildLab/AI-Literacy-CE-ARCE** — all materials, datasets, and lab links
- **Google Colab** — sign in and open a new notebook before lab day
- **Russell & Norvig, Artificial Intelligence: A Modern Approach** — Chapter 1 for the curious (optional)